

Goal-Oriented Deep Fourier Residual Methods

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Abstract

The Deep Fourier Residual (DFR) method minimizes the H^{-1} dual norm of the weak PDE residual, a loss equivalent to the H^1 energy-norm error. In many applications the goal is not the global error but a quantity of interest (QoI): a point value, an average, or a flux. We introduce a goal-oriented DFR method that applies the dual-weighted residual framework to the DFR loss: a primal network and an adjoint network are trained together, the primal minimizing a QoI-weighted residual while the adjoint solves its own DFR problem, so that it approximates the true adjoint rather than merely zeroing the pairing. Our main result is a computable, one-sided bound on the QoI error, verified on every run at about ten times the true error. We also find that the QoI-weighted loss must not be read as an error estimator: training drives it orders of magnitude below the error that remains, and the bound is carried by its remainder. Empirically, the method reallocates accuracy from the global solution to the quantity of interest. Its size must be judged at equal compute, since goal-orientation trains a second network: at a matched training budget the two-dimensional QoI error is 2.3 to 3.9 times smaller, but at matched cost the improvement falls to about a factor of two (paired median 2.2, 95% interval $[1.2, 4.6]$) in the network-size setting and vanishes into the seed noise in a sweep over problem difficulty. The equal-cost benefit is modest and setting-dependent, though genuine; the theory predicts none of it.

Keywords: Deep Fourier Residual, physics-informed neural networks, dual-weighted residual, goal-oriented error estimation, quantity of interest

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1. Introduction

Neural networks have been used to solve differential equations since at least the work of Lagaris et al. [1], their ability to represent solutions resting on classical universal-approximation results [2]. Physics-Informed Neural Networks (PINNs) [3] have since emerged as a popular paradigm for solving partial differential equations (PDEs), embedding physical constraints directly into the neural network training loss.

Taylor, Pardo, and Muga [4] introduced the Deep Fourier Residual (DFR) method, which uses the H^{-1} dual norm of the weak residual as the loss function. This approach provides several advantages over strong-form PINNs:

- The loss is equivalent to the H^1 error for well-posed problems
- Reliable error estimation during training
- Applicability to problems with limited regularity

1.1. Motivation

The DFR loss controls the *global* energy-norm error. In many applications, however, the object of interest is not the global error but a functional of the solution – a quantity of interest (QoI) such as a point value $u(x_0)$, a subdomain average, or a boundary flux. Driving the global error to a fixed tolerance can be wasteful when only the QoI matters, and conversely a small global error does not guarantee an accurate QoI.

The dual-weighted residual (DWR) framework [5] addresses exactly this in the finite element setting by introducing an adjoint solution that localizes the error with respect to the QoI, including the control of pointwise quantities [6]. More recently, machine learning has been used to weight finite-element residuals toward a quantity of interest [7, 8]; those methods are data-driven and require a known target functional, rather than adjoint-based in the DWR sense. Neural networks have also been used inside the DWR framework itself: Roth et al. [9] solve the adjoint problem with a network while keeping a classical finite element primal, and Chakraborty et al. [10] represent both the primal and the adjoint by networks for multigoal estimation. Both estimate the error of a discretization; neither uses the goal to define the training loss, and neither works in the DFR dual-norm setting. We bring DWR to the DFR dual-norm loss: the resulting goal-oriented loss is the QoI-weighted residual functional, evaluated with the same discretization DFR already uses for the H^{-1} norm.

1.2. Contributions

The main contributions of this work are:

1. A goal-oriented DFR method that applies the dual-weighted residual framework to the DFR H^{-1} dual-norm loss, with simultaneous training of a primal and an adjoint network.
2. A goal-oriented counterpart of the DFR error-loss equivalence: a theorem showing that the QoI-weighted dual-norm loss, plus a computable remainder, controls the error $|J(u^*) - J(u_h)|$ in the quantity of interest, where u^* is the exact solution and u_h its approximation. The bound is one-sided, and we delimit what it does and does not claim.
3. An empirical verification of that bound, which holds on every run at a median of ten times the true error. The verification also shows that the QoI-weighted loss alone is not an error estimator: training drives it five orders of magnitude below the error that remains.
4. An empirical study delimiting the mechanism and its size. Goal-orientation reallocates accuracy from the global solution to the quantity of interest, helping wherever plain DFR leaves QoI-relevant residual on the table. Measured at equal computational cost – the fair comparison, since goal-orientation trains a second network – the improvement is about twofold in the two-dimensional network-size setting (paired median 2.2, 95% interval [1.2, 4.6]) and not reliably present in a sweep over problem difficulty; we report the larger matched-budget figures as an upper bound.

1.3. Organization

The remainder of this paper is organized as follows. Section 2 reviews the necessary background on PINNs and the DFR method. Section 3 presents the goal-oriented DFR method. Section 4 provides the theoretical analysis. Section 5 describes our numerical experiments, and Section 6 discusses the results. Finally, Section 7 concludes the paper.

2. Background

2.1. Problem Setting

We consider the abstract problem of finding $u^* \in U$ such that

$$\mathcal{R}(u^*) = 0, \tag{1}$$

where $\mathcal{R} : U \rightarrow V^*$ is the PDE operator mapping from a trial space U to the dual of a test space V .

For example, Poisson’s equation with homogeneous Dirichlet boundary conditions:

$$-\Delta u^* = f \quad \text{in } \Omega, \quad u^* = 0 \quad \text{on } \partial\Omega, \quad (2)$$

can be expressed in weak form as: find $u^* \in H_0^1(\Omega)$ such that

$$\langle \nabla u^*, \nabla v \rangle_{L^2} = \langle f, v \rangle_{L^2} \quad \forall v \in H_0^1(\Omega). \quad (3)$$

2.2. Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) [3] approximate the solution using a neural network u_θ and minimize a loss function based on the PDE residual:

$$\mathcal{L}_{\text{PINN}}(\theta) = \frac{1}{N_f} \sum_{i=1}^{N_f} |\mathcal{R}^s(u_\theta)(x_i)|^2 + \frac{1}{N_u} \sum_{j=1}^{N_u} |u_\theta(x_j) - u^*(x_j)|^2, \quad (4)$$

where $\mathcal{R}^s$ denotes the strong-form residual, $\{x_i\}$ are collocation points, and the second term fits the prescribed initial and boundary data at $\{x_j\}$. Weighting the two terms against each other is a common later refinement.

2.3. Deep Fourier Residual Method

Variational (weak-form) PINNs replace the strong residual by a weak one tested against a chosen set of functions [11], and a posteriori error analyses are available in some settings [12]. Any such method must integrate the weak forms numerically, and the resulting quadrature error is itself a source of inaccuracy that can dominate the discretization [13]. The DFR method [4] takes this further and minimizes the H^{-1} norm of the weak residual:

$$\mathcal{L}_{\text{DFR}}(\theta) = \|\mathcal{R}^w(u_\theta)\|_{V^*}^2. \quad (5)$$

For a box domain with homogeneous Dirichlet conditions, the sine functions ϕ_k are the L^2 -orthonormal eigenfunctions of the Dirichlet Laplacian, with eigenvalues $\lambda_k = \sum_i (\pi k_i / L_i)^2$. Taking the test space $V = H_0^1(\Omega)$ with the energy inner product $\langle u, v \rangle_V = \langle \nabla u, \nabla v \rangle_{L^2}$, the dual norm of the residual is computed from its sine coefficients $\hat{R}_k$ by

$$\|R\|_{V^*}^2 = \sum_k \frac{|\hat{R}_k|^2}{\lambda_k}, \quad (6)$$

which the Discrete Sine Transform evaluates efficiently. In $d \geq 2$ the eigenvalue couples the axes through the *sum* of squared frequencies; it is not the product of the one-dimensional weights.

We depart here from [4] in one respect. That work equips V with the full H^1 inner product, so that its spectral weights carry an additive constant, $\lambda_k = 1 + \sum_i (\pi k_i / L_i)^2$. We use the energy inner product instead. The two norms are equivalent on $H_0^1(\Omega)$ by the Poincaré inequality, so the choice does not affect well-posedness; we prefer the energy inner product because it makes the model problem’s stability constants exactly $\gamma = M = 1$ (Section 4), which removes them from the error bounds. We write $\|\cdot\|_{V^*}$ rather than $\|\cdot\|_{H^{-1}}$ to keep the distinction visible.

In practice the sum in (6) is truncated at a finite mode count K per axis, and the coefficients $\hat{R}_k$ are themselves computed by quadrature. Both approximations are inherited from DFR, and both matter for the error bounds of Section 4: since the truncated sum omits non-negative terms, it *underestimates* the dual norm, so a computed bound is a true bound only insofar as the discarded tail is negligible. We return to this point in Section 4.

2.3.1. Key Properties

The DFR method enjoys the following properties:

Proposition 2.1 (Error equivalence [4]). *For well-posed problems, there exist constants $\gamma, M > 0$ such that, for every $w \in U$,*

$$\frac{1}{M} \|\mathcal{R}(w)\|_{V^*} \leq \|w - u^*\|_U \leq \frac{1}{\gamma} \|\mathcal{R}(w)\|_{V^*}. \quad (7)$$

This error equivalence ensures that minimizing the DFR loss is equivalent to minimizing the error, providing reliable training dynamics.

2.4. From global error to quantities of interest

The error equivalence above controls the *global* energy-norm error $\|w - u^*\|_U$. When the object of interest is a functional J of the solution, this global control is indirect: a small DFR loss guarantees a small H^1 error, but not necessarily an accurate $J(u_\theta)$ at a given computational budget. The dual-weighted residual framework [5] closes this gap by introducing an adjoint solution that relates the QoI error to the residual, which is the starting point for the method developed in Section 3.

3. Methodology

In this section we present the goal-oriented DFR method.

3.1. Overview

Let u^* solve the weak problem $a(u^*, v) = \ell(v)$ for all $v \in V$, with residual $R(u)(v) = \ell(v) - a(u, v)$. The DFR method approximates u^* by a network u_θ and minimizes $\|R(u_\theta)\|_{V^*}^2$. We instead target a quantity of interest $J : U \rightarrow \mathbb{R}$. We introduce the adjoint problem and an adjoint network, and minimize a QoI-weighted residual functional.

3.2. Goal-oriented loss

For a (linear) QoI $J \in U^*$, the adjoint solution $z^* \in V$ solves

$$a(w, z^*) = J(w) \quad \forall w \in U. \quad (8)$$

The error in the QoI admits the representation

$$J(u^*) - J(u_\theta) = a(u^* - u_\theta, z^*) = R(u_\theta)(z^*). \quad (9)$$

Approximating the adjoint by a network z_ϕ , we define the goal-oriented loss

$$\mathcal{L}_{\text{QoI}}(\theta, \phi) = |R(u_\theta)(z_\phi)|, \quad (10)$$

and a regularized objective that also keeps the adjoint network close to the true adjoint,

$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{\text{QoI}}(\theta, \phi) + \alpha \|R^*(z_\phi)\|_{U^*}^2 + \beta \|R(u_\theta)\|_{V^*}^2, \quad (11)$$

where $R^*(z)(w) = J(w) - a(w, z)$, $w \in U$, is the adjoint residual, and $\alpha, \beta \geq 0$ are weights. The dual pairing $R(u_\theta)(z_\phi)$ and the dual norms are evaluated with the DFR discretization (Section 2). Our implementation ties the two weights, $\alpha = \beta$, which is the setting reported throughout; the experiments of Section 5 use $\alpha = \beta = 1$ in one dimension and $\alpha = \beta = 5$ in two, and Section 6.1 reports the 1D control at both.

A subtlety in the joint training deserves emphasis. The goal term $|R(u_\theta)(z_\phi)|$ must update only the *primal* network. If instead the adjoint parameters ϕ were allowed to descend this term, the optimizer could drive the pairing to zero trivially by making z_ϕ orthogonal to the current residual, without z_ϕ approximating z^* at all – a degenerate solution that destroys the localization the method relies on. We therefore hold the adjoint fixed in the goal term (a stop-gradient), so that z_ϕ is trained *solely* by its own DFR residual $\|R^*(z_\phi)\|_{U^*}^2$ and genuinely approximates the true adjoint, while u_θ is guided by the QoI-weighted residual.

The stop-gradient closes the adjoint side of this degeneracy but not the primal side, and it is worth being precise about what the goal term then does.

Since $R(u_\theta)(z_\phi)$ is a single scalar, the primal network can zero it without approximating u^* : the goal term enforces one orthogonality condition, not accuracy. What makes it useful is that $|\cdot|$ has a non-vanishing gradient at the origin, so the term acts as a persistent constraint $R(u_\theta)(z_\phi) \approx 0$ rather than a penalty that switches off once satisfied. This is the surrogate, in a setting with no discrete test space, for the Galerkin orthogonality that a finite element method would enjoy against its whole test space, and by Proposition 4.2 it is exactly what reduces the QoI error to the second-order remainder. Accuracy itself must come from the β term. The two roles are complementary, and the weights are not interchangeable: β buys energy accuracy and the goal term redirects it.

3.3. Evaluation of the dual pairing

For a network u_θ satisfying the boundary conditions, the residual acts on a test function v as $R(u_\theta)(v) = \int_\Omega (f + \Delta u_\theta) v \, dx$. The goal-oriented pairing is therefore the L^2 integral

$$R(u_\theta)(z_\phi) = \int_\Omega (f + \Delta u_\theta) z_\phi \, dx, \quad (12)$$

which we evaluate directly on the same quadrature grid already used to assemble the DFR transforms, so no additional quadrature is introduced. Equivalently, by Parseval’s identity in the orthonormal sine basis, this equals $\sum_k \hat{R}_k \hat{z}_k$; we use the physical-domain form to avoid transforming the adjoint network.

3.4. Algorithm

Algorithm 1 Goal-Oriented DFR

Require: Primal network u_θ , adjoint network z_ϕ , QoI J , weights α, β

Ensure: Trained parameters θ^*, ϕ^*

- 1: Initialize θ, ϕ ; precompute the DST matrix and H^{-1} weights
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Evaluate residual coefficients $\hat{R}(u_\theta)$ and $\hat{R}^*(z_\phi)$ by DST
 - 4: Primal loss $\mathcal{L}_u = |R(u_\theta)(\bar{z}_\phi)| + \beta \|R(u_\theta)\|_{V^*}^2 \triangleright \bar{z}_\phi$: adjoint held fixed
 - 5: Adjoint loss $\mathcal{L}_z = \alpha \|R^*(z_\phi)\|_{U^*}^2$
 - 6: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_u$; $\phi \leftarrow \phi - \eta \nabla_\phi \mathcal{L}_z$
 - 7: **end for**
 - 8: **return** θ^*, ϕ^*
-

3.5. Boundary condition enforcement

For homogeneous Dirichlet conditions, both networks use the cutoff construction

$$u_\theta(x) = \varphi(x) \cdot \tilde{u}_\theta(x), \quad z_\phi(x) = \varphi(x) \cdot \tilde{z}_\phi(x), \quad (13)$$

where φ vanishes on $\partial\Omega$ and $\tilde{u}_\theta, \tilde{z}_\phi$ are the raw network outputs.

4. Theoretical Analysis

We now prove that the goal-oriented loss controls the error in the quantity of interest. This is the goal-oriented counterpart of the DFR error-loss equivalence [4] and is the central theoretical result of the paper. We stress at the outset that it is one-sided: it is a reliability bound, not an equivalence. The DFR estimate (16) bounds the error both above and below, whereas no lower bound holds here. Taking z_h far from z^* makes $\mathcal{L}_{\text{QoI}}$ positive while the QoI error may vanish, so the loss cannot be bounded by the error it controls. All proofs are elementary and given inline.

4.1. Setting and assumptions

Let U and V be Hilbert spaces (trial and test), $a : U \times V \rightarrow \mathbb{R}$ a bilinear form, and $\ell \in V^*$. The primal problem is to find $u^* \in U$ with

$$a(u^*, v) = \ell(v) \quad \forall v \in V. \quad (14)$$

For $w \in U$ define the residual functional $R(w) \in V^*$ by $R(w)(v) = \ell(v) - a(w, v)$. Then $R(u^*) = 0$ and, subtracting (14),

$$R(w)(v) = a(u^* - w, v) \quad \forall v \in V. \quad (15)$$

A word on notation. We write $w \in U$ for an arbitrary trial function, and reserve $u_h \in U$ and $z_h \in V$ for the pair whose quantity-of-interest error the results below control. Nothing in this section requires either to be a network: the trained u_θ and z_ϕ of Section 3 are the case we apply the results to, and the case Sections 5 and 6 measure.

We assume:

(A1) *Boundedness*: $|a(w, v)| \leq M \|w\|_U \|v\|_V$ for all $w \in U, v \in V$.

(A2) *Inf-sup stability*: there is $\gamma > 0$ such that

$$\inf_{0 \neq w \in U} \sup_{0 \neq v \in V} \frac{a(w, v)}{\|w\|_U \|v\|_V} \geq \gamma \quad \text{and} \quad \inf_{0 \neq v \in V} \sup_{0 \neq w \in U} \frac{a(w, v)}{\|w\|_U \|v\|_V} \geq \gamma.$$

Conditions (A1)–(A2) are the hypotheses of the theorem Ern and Guermond [14] name after Banach, Necas and Babuska (BNB), following Necas’ 1962 generalization of the Lax–Milgram theorem and Babuska’s introduction of the inf–sup condition into finite element analysis [15]; see [16] for the Banach-space residual-minimization setting used here. They make the primal problem (14) and its adjoint (below) well posed. For the symmetric coercive model problem (Poisson, $U = V = H_0^1(\Omega)$, $a(u, v) = \langle \nabla u, \nabla v \rangle_{L^2}$, energy norm $\|v\|_U^2 = a(v, v)$) they hold with $\gamma = M = 1$.

Under (A1)–(A2) the DFR error–loss equivalence holds [4]: by (15), (A1), and the first inequality of (A2),

$$\frac{1}{M} \|R(w)\|_{V^*} \leq \|u^* - w\|_U \leq \frac{1}{\gamma} \|R(w)\|_{V^*}, \quad \|R(w)\|_{V^*} := \sup_{0 \neq v \in V} \frac{R(w)(v)}{\|v\|_V}. \quad (16)$$

4.2. Quantity of interest and adjoint

Let the quantity of interest be a bounded linear functional $J \in U^*$. The associated *adjoint problem* is to find $z^* \in V$ such that

$$a(w, z^*) = J(w) \quad \forall w \in U, \quad (17)$$

which is well posed by (A1) and the second inequality of (A2). For $z \in V$ define the adjoint residual $R^*(z) \in U^*$ by $R^*(z)(w) = J(w) - a(w, z)$. Then $R^*(z^*) = 0$ and, subtracting (17),

$$R^*(z)(w) = a(w, z^* - z) \quad \forall w \in U. \quad (18)$$

Remark 4.1. For $d \geq 2$ the point-evaluation functional $u \mapsto u(x_0)$ is not bounded on H^1 . We realize it by the mollification $J_\sigma(u) = \int_\Omega \eta_\sigma(x - x_0) u(x) dx$ with η_σ a normalized Gaussian of width σ ; its Riesz representative η_σ lies in $L^2(\Omega) \subset H^{-1}(\Omega) = U^*$, so $J_\sigma \in U^*$, and $J_\sigma \rightarrow \delta_{x_0}$ as $\sigma \rightarrow 0$. In one dimension $H^1(\Omega) \hookrightarrow C^0(\bar{\Omega})$, so point evaluation is already bounded and mollification is not needed on theoretical grounds. We nevertheless mollify in one dimension as well, at the same σ used for the adjoint right-hand side, so that the 1D and 2D experiments measure the same functional and the 1D control is comparable to the 2D result.

4.3. Error representation

Proposition 4.2 (Exact error representation). *Let u^* solve (14) and z^* solve (17). For every $u_h \in U$ and every $z_h \in V$,*

$$J(u^*) - J(u_h) = R(u_h)(z^*) = R(u_h)(z_h) + R(u_h)(z^* - z_h). \quad (19)$$

Proof. Taking $w = u^* - u_h$ in (17) and then using (15),

$$J(u^*) - J(u_h) = J(u^* - u_h) = a(u^* - u_h, z^*) = R(u_h)(z^*).$$

The second equality in (19) is the splitting $z^* = z_h + (z^* - z_h)$ together with the linearity of $R(u_h)$. $\square$

The first term $R(u_h)(z_h)$ is computable from the two networks; it is the goal-oriented loss

$$\mathcal{L}_{\text{QoI}}(u_h, z_h) := |R(u_h)(z_h)|. \quad (20)$$

The second term involves the unknown exact adjoint z^* and is bounded next.

4.4. Goal-oriented error control

Theorem 4.3 (QoI error control). *Under (A1)–(A2), for every $u_h \in U$ and $z_h \in V$,*

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{1}{\gamma} \|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}. \quad (21)$$

Proof. By Proposition 4.2, (20), and the triangle inequality,

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + |R(u_h)(z^* - z_h)|.$$

For the remainder, the definition of the dual norm gives

$$|R(u_h)(z^* - z_h)| \leq \|R(u_h)\|_{V^*} \|z^* - z_h\|_V.$$

It remains to control $\|z^* - z_h\|_V$ by the adjoint residual. Applying the second inf-sup inequality of (A2) to $z^* - z_h \in V$ and using (18),

$$\gamma \|z^* - z_h\|_V \leq \sup_{0 \neq w \in U} \frac{a(w, z^* - z_h)}{\|w\|_U} = \sup_{0 \neq w \in U} \frac{R^*(z_h)(w)}{\|w\|_U} = \|R^*(z_h)\|_{U^*}.$$

Combining the three displays yields (21). $\square$

The three quantities on the right-hand side of (21) are exactly the terms minimized in the training objective of Section 3: the goal-oriented loss $\mathcal{L}_{\text{QoI}} = |R(u_h)(z_h)|$ and the primal and adjoint DFR dual norms $\|R(u_h)\|_{V^*}$ and $\|R^*(z_h)\|_{U^*}$, which DFR evaluates spectrally. The bound (21) is therefore a posteriori and computable from the two trained networks alone, and the goal-oriented objective is the quantity it controls.

Two caveats attach to the word computable. First, the dual norms are spectral sums truncated at K modes per axis and assembled by quadrature

(Section 2); the truncated sum discards non-negative terms, so it underestimates $\|R(u_h)\|_{V^*}$, and (21) is a guaranteed bound only up to the discarded tail. It is a true bound under the standard saturation assumption that the tail is a fixed fraction of the total, which the mode counts used in Section 5 make reasonable but which we do not verify. Second, and more consequentially in practice, $\mathcal{L}_{\text{QoI}}$ is itself minimized during training. We return to what that does to (21) in Remark 4.5 and measure it in Section 6.6.

Corollary 4.4 (Adjoint trained to tolerance). *If the adjoint network satisfies $\|R^*(z_h)\|_{U^*} \leq \varepsilon$, then*

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{\varepsilon}{\gamma} \|R(u_h)\|_{V^*} \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{M\varepsilon}{\gamma} \|u^* - u_h\|_U,$$

the last step by the left inequality of (16), which bounds $\|R(u_h)\|_{V^}$ by $M \|u^* - u_h\|_U$.*

Remark 4.5 (Second-order structure, and what it does not say). *The remainder in (21) is the product of the primal and adjoint dual-norm residuals. If both networks are trained to energy accuracy $O(\delta)$, that is $\|R(u_h)\|_{V^*}, \|R^*(z_h)\|_{U^*} = O(\delta)$, then the remainder is $O(\delta^2)$: driving the goal term to zero leaves a QoI error that is quadratic in the energy accuracy of the two networks. This product structure is the characteristic feature of dual-weighted residual methods.*

Four qualifications matter, and we state them because the natural reading of the previous paragraph overreaches on all four.

First, the $O(\delta^2)$ statement presupposes that δ is small, that is, that both networks are well resolved in the energy norm. It therefore says nothing about a regime in which global resolution is out of reach: there δ is not small, and $\gamma^{-1}\delta^2$ is not small either. The bound is weakest exactly where a goal-oriented method is most wanted, and it does not by itself predict an advantage there. Section 6.3 finds the advantage there anyway, so the bound accommodates the result without explaining it.

Second, the classical second-order gain rests on Galerkin orthogonality, which annihilates the first-order term $R(u_h)(z_h)$ for every discrete test function. Here there is no discrete test space and no such orthogonality. Minimizing $\mathcal{L}_{\text{QoI}}$ enforces a single scalar condition, $R(u_h)(z_h) \approx 0$, in place of orthogonality against a whole subspace. That is enough for the error representation to reduce to the second-order remainder, but it is a far weaker constraint, and it is satisfiable without u_h being accurate.

Third, the entire analysis is approximation-theoretic and contains no model of the optimization. This is a real limitation rather than a formality: Section 6.4 finds that plain DFR is often far from convergence at a practical

training budget, so its residual reflects how far training has progressed and not only what its discretization can represent. Nothing in (21) distinguishes a residual that is large because the network cannot do better from one that is large because training has not yet finished, and the two call for different remedies.

Fourth, and consequently, $\mathcal{L}_{\text{QoI}}$ must not be read as an error estimator for a network trained on it. The optimizer minimizes it directly; being one scalar condition, it is driven essentially to zero regardless of the error that remains. Section 6.6 measures this: $\mathcal{L}_{\text{QoI}}$ settles around 10^{-10} while the QoI error sits near 10^{-4} , some six orders of magnitude apart. A quantity minimized to zero cannot report what error is left. The useful computable object is the whole right-hand side of (21), which is carried by the remainder term and which we verify there.

Remark 4.6 (Recovery of standard DFR). *Taking the supremum of the error representation over all unit goals recovers global energy control: since $\sup_{\|J\|_{U^*}=1} J(u^* - u_h) = \|u^* - u_h\|_U$, the worst case over goals reproduces the DFR equivalence (16). Standard DFR therefore controls every quantity of interest simultaneously, whereas goal-oriented DFR controls a single prescribed goal, which is why the latter can be cheaper for a fixed target accuracy.*

5. Numerical Experiments

We study the goal-oriented method against a plain DFR baseline on one- and two-dimensional Poisson problems, asking whether the QoI-weighted loss yields a more accurate quantity of interest at matched cost.

5.1. Setup

5.1.1. Problems

In one dimension we use two manufactured solutions with homogeneous Dirichlet conditions:

- **Smooth:** $u(x) = \sin(2x)$ on $(0, \pi)$, with $f = 4 \sin(2x)$.
- **Peaked:** an arctan profile of steepness $k = 8$ on $(0, 1)$, $u(x) = \arctan(k(x - \frac{1}{2})) - \ell(x)$ with ℓ the affine correction enforcing $u(0) = u(1) = 0$; its gradient is concentrated near $x = \frac{1}{2}$.

In two dimensions we use a separable bump on the unit square $(0, 1)^2$, $u(x, y) = A(x)A(y)$ with A the boundary-corrected arctan profile of steepness $k = 8$ above; its gradient is concentrated along the lines $x = \frac{1}{2}$ and $y = \frac{1}{2}$, and $f = -\Delta u$ follows in closed form.

We use the word peaked rather than sharp advisedly. At $k = 8$ the profile’s gradient varies by only a factor of about three between its peak and its median, and the same $k = 8$ profile appears in the 1D problem, where it is resolved without difficulty. The 2D solution is not intrinsically harder than the 1D one; what changes is the dimension, and with it the difficulty of the optimization at a given network size. We return to this in Section 6.7. The quantity of interest is a point evaluation $J(u) = u(x_0)$, mollified by a normalized Gaussian of width σ (Section 4): at $x_0 = \pi/4$ (1D smooth) and $x_0 = 0.65$ (1D sharp), and at $x_0 = (0.65, 0.65)$ in 2D. The width σ balances localization against spectral resolution: too small a σ makes J_σ approach a Dirac spike, whose high-frequency content the truncated sine basis cannot represent without a large mode count, while too large a σ blurs the target away from a point value. We use $\sigma = 0.04$ in 1D and $\sigma = 0.07$ in 2D, which resolve well at the mode counts employed here.

For reference, the 1D experiments use 60 sine modes and 150 quadrature points, 2000 Adam steps at learning rate 2×10^{-3} , and 400 LBFGS iterations; the 2D experiments use 16 modes and 32 quadrature points per axis, 3000 Adam steps at learning rate 3×10^{-3} , and 500 LBFGS iterations. The mode budgets are not the limiting factor in either case: the $k = 8$ profile’s sine coefficients decay fast enough that 16 modes already capture all but a fraction 10^{-7} of its energy, and 60 modes all but 10^{-11} .

5.1.2. Varying the difficulty

Comparing one dimension against two confounds the difficulty of the problem with everything else that differs between the two settings, including the mode count, the quadrature and the step budget. Our principal experiment therefore varies the difficulty *within* a single setting: we sweep the steepness $k \in \{2, 4, 8, 16\}$ of the 2D bump at two fixed network sizes (105 and 337 primal parameters) and one fixed discretization, over ten seeds, so that k is the only variable.

The mode count for this sweep is fixed at 24 per axis with 48 quadrature points, chosen for the sharpest case rather than inherited from the sweep above. This matters: the relative energy the sine basis fails to represent grows from 8×10^{-8} at $k = 8$ to 10^{-3} at $k = 32$, so a sweep that held the mode count at 16 would make the test space, and not the network, the binding constraint at large k , and would measure mode truncation while appearing to measure resolution. At 24 modes the tail stays below 7×10^{-7} for every k in the sweep. We use two network sizes rather than one because the smaller is the more resolution-limited and the more favorable to goal-orientation, while the larger is where the sweep of Section 6.2 finds goal-orientation weakest; a

behavior present at both cannot be an artifact of the size we chose.

5.1.3. *Methods and training*

The plain DFR baseline trains a single network against the H^{-1} residual loss. The goal-oriented method trains a primal and an adjoint network against the objective of Section 3. Both use the same primal-network architecture, the same Fourier test space, and the same schedule: Adam followed by an LBFGS polish, at the same step counts and learning rate. In two dimensions both methods use the identical full tensor-product DST discretization of the weak residual, so the comparison isolates exactly the effect of adding goal-orientation. The cutoff construction enforces the boundary conditions. We sweep the network width and repeat over ten random seeds in two dimensions and three in one.

Degrees of freedom throughout count the *primal* network. The goal-oriented method also trains an adjoint network of the same size, so at a given abscissa it carries about twice the parameters and twice the per-step cost of the baseline. We match on the primal network because the adjoint is auxiliary and is discarded once the primal is trained, but the comparison is not one of equal total cost, and we make no claim that it is.

As a correctness check, we verify that the discrete DFR loss vanishes at the exact solution of each of the three problems. The check is by construction independent of each problem’s closed-form forcing: the Laplacian of the exact solution is taken by finite differences, so the test detects a mismatch between the manufactured solution and its forcing rather than reproducing it. On all three problems the loss at u^* falls at least eight orders of magnitude below its value at a one-percent perturbation of u^* , and it increases monotonically with the size of that perturbation.

5.1.4. *Metrics*

We report the error in the (mollified) quantity of interest, $|J_\sigma(u_h) - J_\sigma(u^*)|$, as a function of the primal-network degrees of freedom. Results and their interpretation are given in Section 6.

5.1.5. *Reproducibility*

Code is available in the accompanying repository, <https://github.com/caverac/hp-dfr>, with documentation at <https://caverac.github.io/hp-dfr/>; the figures are regenerated from saved sweep data by a single command.

6. Results and Discussion

We compare goal-oriented DFR against a plain DFR baseline throughout, at matched primal-network degrees of freedom. The metric is the error in the mollified point QoI, $|J_\sigma(u_h) - J_\sigma(u^*)|$, reported as a median over seeds with bands spanning the seed min–max.

Sections 6.1 and 6.2 sweep the network width in one and two dimensions, and Section 6.3 varies the problem difficulty directly within one setting; all three are at a matched training budget. Section 6.4 then measures what goal-orientation does to the global solution and finds that plain DFR is not converged at that budget, which motivates Section 6.5, the comparison at matched *cost* that we take as primary. Section 6.6 tests the theorem against the errors actually committed. The picture is one mechanism: goal-orientation reallocates accuracy from the global solution to the quantity of interest, which helps wherever plain DFR leaves QoI-relevant residual unresolved. At matched budget that help looks large; at equal cost it is about twofold in two dimensions and unreliable in the steepness sweep.

6.1. One dimension: the baseline is already accurate at the QoI

Figure 1 reports the 1D results on the smooth and peaked problems. Plain DFR reaches a QoI error of about 3×10^{-5} already at the smallest networks, a few tens of parameters, and does not improve on it as the network grows: on the smooth problem the curve is flat to within a factor of two across the whole sweep, and if anything drifts upward with width. We took this at first for saturation, the residual being driven near zero so that the error–loss equivalence (16) makes the QoI accurate everywhere. Section 6.4 shows the reading is not quite right – the baseline is not converged even here – but the practical conclusion for one dimension stands: plain DFR is already accurate at the QoI, and there is little for goal-orientation to add.

With the baseline already accurate at the QoI, goal-orientation has little to add, and at small networks it pays for the attempt. At $\alpha = \beta = 1$ it is 3.5 to 21 times *less* accurate than plain DFR at small and moderate sizes, where its two-network objective is harder to optimize, and it draws level only at the largest.

Because the 2D experiments below use a different loss weight, the 1D sweep is run at both. Raising the weight to $\alpha = \beta = 5$ improves the goal-oriented QoI error by up to six times, but it does not overturn the picture: goal-orientation wins 7 of 24 runs at $\alpha = \beta = 5$ against 4 of 24 at $\alpha = \beta = 1$, and loses the median at every size but the largest. The weight alone does

not manufacture an advantage, which is what licenses comparing the two dimensions at all.

One detail here is easy to dismiss as noise and turns out not to be. At the largest network, $\alpha = \beta = 5$ does not merely reach parity but *overtakes* plain DFR on both problems, by 5 times on the smooth one and 2 on the peaked one. We flagged this initially as an oddity; it is in fact the same reallocation the 2D experiments show, appearing wherever the baseline has QoI-relevant residual left over – here because, as Section 6.4 shows, the baseline is not fully converged even on these easy 1D problems. One dimension is therefore a weaker control than it first appears: it shows goal-orientation does not manufacture an advantage at small networks, but not the absence of one everywhere.

6.2. Two dimensions: a goal-oriented advantage

We repeat the comparison in two dimensions on the separable bump of Section 5. Both methods use the identical full tensor-product DST discretization and the identical training budget, so the comparison isolates exactly the effect of adding goal-orientation. Plain DFR sits at a median QoI error of 1.5×10^{-4} at the smallest networks and improves to around 10^{-5} as the width grows. This solution is harder to resolve than the 1D ones, and, as Section 6.4 shows, the baseline does not fully converge on it within the shared budget.

Figure 2 reports the outcome, and Table 1 gives the medians over ten seeds. Goal-orientation is the more accurate method at every size, winning 33 of the 40 runs. The median improvement is 2.3 to 3.9 times, with a geometric mean of 3.1, and it is remarkably flat across a twelvefold range of network sizes. Pooling the per-run paired ratios gives a median of 3.1 with a bootstrap 95% confidence interval of $[2.5, 6.8]$, so the effect is comfortably separated from unity in aggregate.

Ten seeds are necessary rather than decorative here, and it is worth saying why. The seed-to-seed spread of plain DFR at fixed size reaches a factor of 68; a three-seed median is a single middle draw from that, and the size-by-size ordering it produces is not stable. Our own earlier three-seed sweep put the 337 column at near-parity ($1.2\times$) and the 105 column at $5.3\times$, and both were sampling artifacts: at ten seeds they are $2.8\times$ and $3.9\times$. We report the per-size confidence intervals in the interest of not repeating the mistake: at 105, 697 and 1185 they exclude unity ($[2.8, 15.7]$, $[2.7, 18.0]$ and $[1.6, 2.9]$ respectively), whereas at 337 the interval is $[0.7, 19.2]$ and does not. No individual column of this table should be read as establishing a size-dependent trend; the aggregate should.

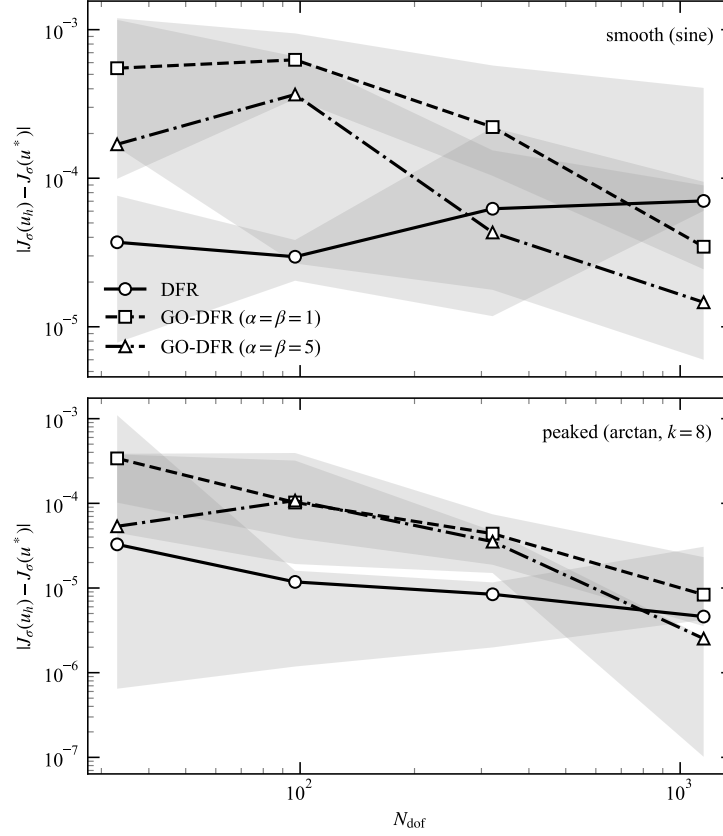


Figure 1: QoI error versus primal-network degrees of freedom (DOF) on 1D Poisson problems (smooth, top; peaked, bottom). Plain DFR reaches a QoI error of about 3×10^{-5} at extremely small network sizes and does not improve on it as the network grows. Goal-oriented DFR (GO-DFR) is the worse method at small and moderate sizes, at either loss weight, so the weight used by the 2D experiments ($\alpha = \beta = 5$) does not by itself manufacture an advantage. At the largest networks $\alpha = \beta = 5$ overtakes the baseline on both problems, the same reallocation the 2D experiments show. Markers show medians over random seeds; bands span the seed min-max.

Neither method is monotone in the degrees of freedom, plain DFR being at its best at 697 and worse at 1185. Given the seed spread we do not attribute this to a mechanism either.

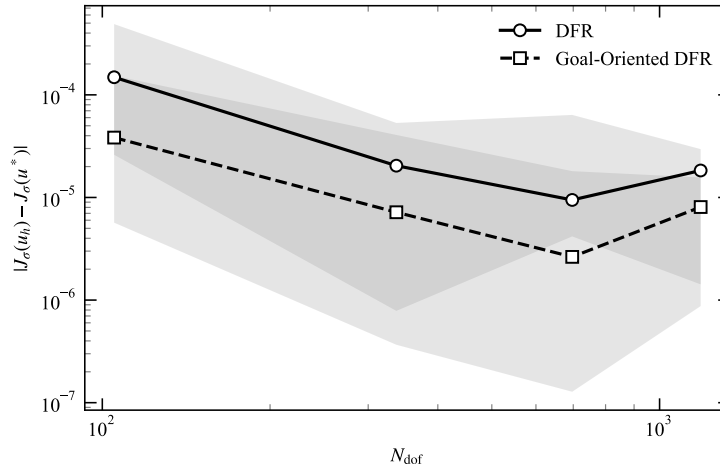


Figure 2: QoI error versus primal-network degrees of freedom (DOF) on the 2D Poisson problem with an arctan bump, at a matched training budget. The goal-oriented DFR method (GO-DFR), which holds the adjoint fixed in the goal term and weights the dual norm by the coupled eigenvalues $\lambda_k = \sum_i (\pi k_i / L_i)^2$, redirects the network’s accuracy toward the support of the QoI. It wins 33 of the 40 runs and reduces the median QoI error by 2.3 to 3.9 times at matched primal-network DOF, a geometric mean of 3.1 times. These are matched-*budget* figures and an upper bound: the baseline is not converged at this budget (Section 6.4), and GO-DFR trains a second network, so at matched cost the advantage roughly halves (Table 2). Markers show medians over random seeds; bands span the seed min–max.

6.3. Varying the difficulty directly

The sweeps so far vary the network size at a fixed problem. They show *that* goal-orientation helps in two dimensions, but not that it helps *because* the network is resolution-limited: network size and problem difficulty are confounded in a single comparison against the 1D case. The steepness k separates them. Increasing k makes the same solution harder to represent at a fixed network and a fixed discretization, so it is the variable the mechanism is supposed to respond to, and it is the one we vary here. The mode count is held at 24 throughout, high enough that the sine basis captures even the sharpest case to a relative energy tail of 7×10^{-7} ; holding it at the 16 of the previous sweep would have made the test space rather than the network

Table 1: Median point-QoI error on the 2D arctan bump, over ten random seeds, at matched primal-network degrees of freedom (DOF). Goal-oriented DFR (GO-DFR) improves the median QoI error at every size, by a geometric mean of 3.1 times, and wins 33 of the 40 runs. The improvement is flat across a twelvefold range of network sizes; given a seed-to-seed spread reaching a factor of 68, we do not read the size-to-size variation as a trend.

DOF	Plain DFR	GO-DFR	Improvement	GO-DFR wins
105	1.5×10^{-4}	3.8×10^{-5}	$3.9\times$	8/10
337	2.0×10^{-5}	7.2×10^{-6}	$2.8\times$	7/10
697	9.5×10^{-6}	2.6×10^{-6}	$3.6\times$	9/10
1185	1.8×10^{-5}	8.1×10^{-6}	$2.3\times$	9/10

the binding constraint at large k , and the experiment would have measured mode truncation instead. Figure 3 reports the result at two network sizes, ten seeds each.

The prediction is borne out, and the mechanism is visible in the curves rather than in any ratio. Plain DFR changes little while the problem is easy and then degrades sharply: at 105 degrees of freedom its median error grows by a factor of 84 between $k = 2$ and $k = 16$. Goal-oriented DFR degrades by a factor of 11 over the same range. The two curves cross near $k \approx 5$, and beyond it goal-orientation wins by 5.9 times at $k = 8$ and 11.1 at $k = 16$ (9/10 runs each, bootstrap 95% intervals $[2.2, 17.3]$ and $[6.3, 52.8]$). At 337 degrees of freedom the same hardening produces a 15.4-fold advantage at $k = 16$. As the problem becomes harder to represent, plain DFR leaves progressively more residual unresolved, and goal-orientation converts it into QoI accuracy.

The sweep also shows the method can *hurt*, which the network-size sweeps never made visible. At 105 degrees of freedom and $k = 4$ goal-orientation is 2.4 times *worse* than the baseline and wins only 2 of 10 runs: the problem is easy enough that there is no misallocated capacity to recover, and the second network makes the optimization harder for nothing. A practitioner choosing between the two methods needs this fact, and we would not have it from a sweep that varied only the network size.

The surprise is at the other end. At 337 degrees of freedom and $k = 2$ – the *easiest* problem in the study – goal-orientation is still 15.4 times more accurate, winning 9 of 10 runs with a bootstrap interval of $[4.6, 40.0]$. This is not resolution limitation: at that network size plain DFR’s median error is nearly constant across $k = 2, 4$ and 8 ($1.14, 1.48$ and 1.29×10^{-5}), so it is not the sharpness of the solution that limits it. We took this at first for an

accuracy floor. It is not that either. Section 6.4 shows the baseline is simply not converged at the shared training budget: given more steps its error keeps falling. Goal-orientation reaches 7.4×10^{-7} within the shared budget by spending it on the QoI rather than the global solution. The advantage at $k = 2$ is therefore driven by the baseline’s under-convergence at the shared budget, the same effect as the 1D overtake of Section 6.1. We had thought the resolution-limited advantage at large k different in kind, resting on the problem rather than the budget; Section 6.5 shows it is largely not, closing at equal cost as well.

So at matched budget the sweep shows a clear resolution-limited advantage. Whether it survives at equal cost is a separate question, taken up in Section 6.5; Section 6.4 first measures the mechanism directly, and Section 6.7 interprets it.

6.4. *What goal-orientation trades, and against what baseline*

The experiments above report the QoI error alone. Two further quantities settle what the method is doing and against what it is measured: the global energy error of each solution, and how far plain DFR is from convergence at the training budget the comparison uses. We record both on the $k = 2$, 337-parameter problem, where goal-orientation’s advantage is largest and least expected.

Goal-orientation reallocates accuracy. At the shared training budget, and on every one of the three seeds, goal-oriented DFR reaches a smaller QoI error and a *larger* global error. The median global relative L^2 error is 9.8% against plain DFR’s 0.9%, roughly an order of magnitude, and the dual-norm residual is several times larger, while the QoI error is smaller by the fifteen-fold margin the ten-seed sweep of Section 6.3 reports for this configuration. The method does not solve the problem better; it solves a different problem, spending the network on the neighborhood of the quantity of interest at the expense of everywhere else. This is the mechanism the construction was designed to produce, now measured rather than argued, and it carries a warning for use: a goal-oriented solution is trustworthy at its quantity of interest and nowhere else.

The baseline is not converged. Training plain DFR far past the shared budget shows its error still falling. On the 2D problem of Section 6.2 the dual-norm residual drops by a factor of 6 between the 3000 Adam steps used throughout and 50 000, and is still decreasing at the end, with no plateau; on the easy $k = 2$ problem a longer run lowers the QoI by a further factor of about 6. What looked in Sections 6.1 and 6.3 like an accuracy floor – an error insensitive to problem difficulty and network size – is instead a fixed

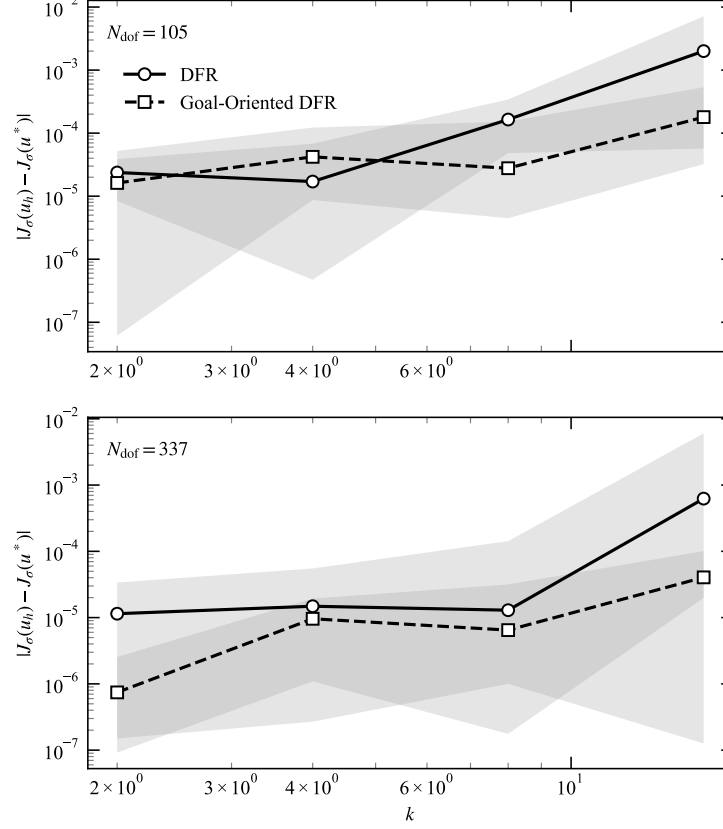


Figure 3: QoI error versus the solution’s steepness k at fixed network size (105 degrees of freedom, top; 337, bottom) and fixed discretization, so that k is the only variable, at a matched training budget. Plain DFR changes little while the problem is easy and then degrades sharply, by a factor of 84 between $k = 2$ and $k = 16$ at 105 degrees of freedom; goal-oriented DFR degrades by a factor of 11 over the same range, so the gap opens as the problem hardens and the two curves cross near $k \approx 5$. Below the crossover, at $k = 4$ and 105 degrees of freedom, goal-orientation is the worse method. At 337 degrees of freedom it is ahead everywhere, including at $k = 2$; there the advantage comes from the baseline being under-converged at the shared budget (Section 6.4) rather than from resolution limitation. This is a matched-budget comparison; at equal cost most of the advantage closes (Section 6.5). Markers show medians over ten random seeds; bands span the seed min-max.

training budget cutting a slow decay at the same point each time. There is no floor; there is a schedule.

This places an honest boundary on the results. The comparisons above are at a matched training *budget*, not at matched accuracy, and the baseline has not converged at that budget. A matched-budget comparison is nonetheless the right first benchmark, because holding the optimizer, step count, learning rate and schedule identical isolates the effect of the goal-oriented objective from any difference in training dynamics, and that is what establishes the reallocation as a property of the loss. But since goal-orientation trains a second network, an equal-budget comparison gives it more compute, so the *size* of the advantage must be read against a comparison at equal cost. We turn to that next.

6.5. At matched cost

We rerun the two-dimensional sweeps giving plain DFR 2.5 times the Adam and LBFGS budget of goal-oriented DFR, so a plain-DFR run costs about the same wall-clock as a goal-oriented run. If anything the baseline is marginally over-resourced: two networks cost roughly twice a single one per step, and $2.5 > 2$, so the comparison is conservative toward plain DFR. All improvements below are paired per-run ratios (each seed compared against itself), which is the appropriate statistic given the large seed-to-seed spread; medians and bootstrap 95% intervals are over ten seeds.

The network-size sweep survives. Table 2 gives the medians. Goal-orientation wins 28 of the 40 runs, and the paired improvement has a median of 2.2 with a confidence interval of $[1.2, 4.6]$, clear of unity. This is roughly half the matched-budget value (a geometric mean of 3.1 became 2.2), but it is real: at equal compute, goal-orientation still buys about a twofold reduction in the point-QoI error on this problem. The one size at which the median plain-DFR run is the more accurate (105 degrees of freedom) is also one at which goal-orientation wins 7 of 10 seeds, a discrepancy that reflects a right-skewed seed distribution rather than a genuine reversal, and is exactly why we report the paired statistic.

The steepness sweep does not survive. Rerun at matched cost, goal-orientation wins only 38 of the 80 runs, and the paired improvement has a median of 0.92 with a confidence interval of $[0.69, 1.85]$ that straddles unity: in aggregate the two methods are now indistinguishable. Restricting to the resolution-limited end ($k \geq 8$), where the matched-budget advantage was largest, leaves a median of 1.6 over 40 runs, but with an interval of $[0.8, 3.1]$ that still includes unity. The several-fold advantages of Section 6.3

were in large part the baseline being under-resourced; once it is trained to comparable cost, most of them close.

We therefore take the matched-cost comparison as the primary one and the matched-budget figures of Sections 6.2 and 6.3 as an upper bound. The result is modest but genuine: goal-orientation reduces the point-QoI error by about a factor of two at equal cost in the two-dimensional network-size setting, and by no reliable margin in the steepness setting. The reallocation mechanism of Section 6.4 is intact – it is a property of the loss, established where training dynamics are held fixed – but its practical payoff at equal compute is smaller, and more setting-dependent, than the matched-budget numbers suggest.

Table 2: Median point-QoI error on the 2D arctan bump at matched *cost*, with plain DFR given $2.5\times$ the budget of goal-oriented DFR (GO-DFR), over ten random seeds. GO-DFR wins 28 of the 40 runs; the aggregate paired improvement is $2.2\times$ with a bootstrap 95% interval of $[1.2, 4.6]$, roughly half the matched-budget value of Table 1. At 105 DOF the median crosses while GO-DFR still wins most seeds (a skewed seed distribution), which is why the headline uses the paired statistic.

DOF	Plain DFR	GO-DFR	GO-DFR wins
105	2.8×10^{-5}	3.8×10^{-5}	7/10
337	1.5×10^{-5}	7.2×10^{-6}	8/10
697	5.5×10^{-6}	2.1×10^{-6}	7/10
1185	9.0×10^{-6}	8.1×10^{-6}	6/10

6.6. Does the bound hold, and is the loss an estimator?

Theorem 4.3 is a posteriori and computable, so it can be checked rather than merely asserted. Every term on the right-hand side of (21) is evaluated at every training step; we record all three at the end of training for each of the 168 goal-oriented runs of the sweeps above (48 in one dimension, 40 in two, and 80 in the steepness sweep), and compare their sum against the QoI error actually committed. Recall $\gamma = 1$ for this problem, so the bound is $\mathcal{L}_{\text{QoI}} + \|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}$. Figure 4 shows the outcome.

Two things stand out, and they pull in opposite directions.

The bound holds, and is usefully tight. It is satisfied on all 168 runs, with no violation anywhere in the study, which spans an eightfold range of problem difficulty and a twelvefold range of network size. This is not a foregone conclusion: as noted in Section 4, the dual norms are evaluated on truncated spectral sums, which underestimate the true norms, so the

computed right-hand side could in principle fall below the error. It does not, and by a comfortable margin: the bound exceeds the true QoI error by a median factor of 15, and never falls below 2.7. Within an order of magnitude, and never optimistic, is a useful a posteriori bound.

The goal-oriented loss is not an error estimator. The same data refutes the natural reading of $\mathcal{L}_{\text{QoI}}$ as an estimate of the QoI error, and it is worth being blunt about it because the reading is inviting. Across all 168 runs $\mathcal{L}_{\text{QoI}}$ has a median value of 2×10^{-10} , against a median QoI error near 10^{-5} : it is smaller than the quantity it is supposed to estimate by roughly five orders of magnitude, and it never once exceeds it. The reason is Remark 4.5: training minimizes $\mathcal{L}_{\text{QoI}}$ directly, and being a single scalar condition it can be driven to zero without u_h being accurate. The optimizer does exactly that. Every part of the bound’s value therefore comes from the remainder $\|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}$, whose median is 1.3×10^{-3} in one dimension.

This has a practical consequence worth stating plainly. A reader who took $\mathcal{L}_{\text{QoI}}$ from a trained goal-oriented network and reported it as an error estimate would understate the error by five orders of magnitude. The estimator to use is the whole right-hand side of (21), and it is the remainder, not the goal term, that carries it. This is a general hazard of training on an a posteriori quantity, and not specific to DFR: any residual functional that appears in the loss is disqualified from also serving as an unbiased estimate of what the loss failed to remove.

6.7. Why the contrast, and what it means

A single mechanism accounts for the whole picture. Goal-orientation reallocates the network’s accuracy from the global solution to the quantity of interest (Section 6.4). It therefore helps precisely to the extent that plain DFR leaves QoI-relevant residual unresolved, and hurts when it does not. The residual can be left unresolved for two reasons that look different but drive the same behavior: the solution may be too sharp for the network to represent at any budget (large k), or the shared training budget may simply be too small to converge an otherwise capable network ($k = 2$ at 337 parameters, and the easy 1D problems). Both leave residual on the table, and goal-orientation converts it into QoI accuracy either way. Where it loses – small networks on easy problems, most sharply $k = 4$ at 105 degrees of freedom – is where plain DFR has little QoI-relevant residual to give up, so the trade returns nothing and the second network is a pure cost. The naive summary “goal-orientation helps when the problem is hard” is wrong at both ends of Figure 3: it fails on the hard-but-small case where the method loses, and it misses the easy-but-under-trained case where the method wins.

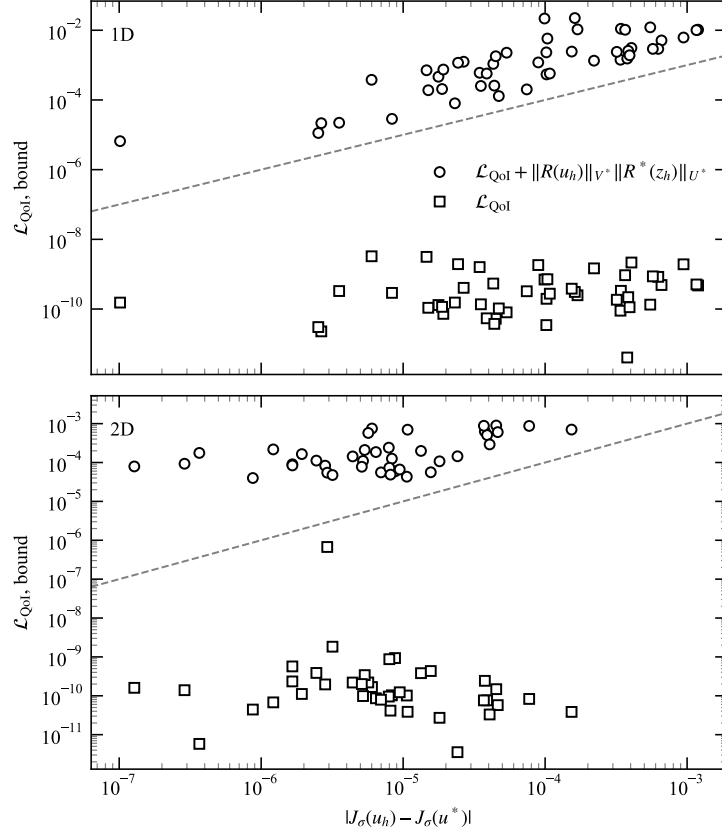


Figure 4: Verification of the QoI error bound (21) on the goal-oriented runs of the network-size sweeps (1D, top; 2D, bottom). The abscissa is the QoI error actually committed; the dashed diagonal is equality. The bound (circles) lies above the diagonal on every run, exceeding the true error by a median factor of 9.8 in 1D and 17.7 in 2D; across all 168 runs of the study, including the steepness sweep, it holds without exception. The goal term $\mathcal{L}_{\text{QoI}}$ alone (squares) lies far *below* the diagonal: training minimizes it directly, driving it some five orders of magnitude beneath the error that remains, so it cannot serve as an error estimator and the bound is carried entirely by its remainder.

We had expected the two reasons to be differently robust: the residual left by a genuinely unrepresentable solution should persist at any budget, whereas that of an under-trained network should not. The matched-cost comparison (Section 6.5) shows the distinction matters less than we thought. Even in the resolution-limited regime, giving plain DFR a proportionally larger budget closes most of the gap: the QoI error it leaves at a fixed budget is, in large part, training that has not finished rather than resolution it cannot reach. What survives at equal cost is the two-dimensional network-size advantage, not the steepness one.

We stress that Theorem 4.3 predicts none of this. It does not predict the resolution-limited gain, for the reason set out in Remark 4.5: the second-order remainder is small only when both networks are well resolved in energy, which is exactly what fails where that gain appears, so the bound is weakest where the method is most useful. It has still less to say about the budget-limited case, which is a statement about how far training has progressed, while our analysis is entirely approximation-theoretic and contains no model of the optimization at all. The mechanism we can point to is the one described in Section 3, that the goal term acts as a persistent orthogonality constraint redirecting accuracy toward the QoI, and no bound we prove quantifies it. Producing a theory that predicts rather than accommodates these results, and one that distinguishes an unrepresentable residual from an untrained one, is the most interesting problem this work leaves open.

Two design points were essential to observing any of this. First, the adjoint network must be trained by its own residual and held fixed in the goal term (Section 3); otherwise it degenerates and the localization signal is lost. Second, in two dimensions the dual-norm weights must use the true eigenvalues $\lambda_k = \sum_i (\pi k_i / L_i)^2$ rather than a separable product of one-dimensional weights, so that high-frequency residuals are penalized correctly.

6.8. Limitations

- The analysis assumes a linear QoI and a bounded, inf-sup-stable bilinear form.
- Theorem 4.3 is a reliability bound only. We prove no efficiency bound, and none holds without further assumptions on z_h , so the loss cannot be used to certify that the QoI error is large.
- The bound is stated for exact dual norms but evaluated on truncated spectral sums assembled by quadrature. It is therefore guaranteed only up to the discarded tail, which we argue is small at the mode counts used but do not bound rigorously.

- The seed-to-seed spread is large, reaching a factor of 68 at fixed network size, so individual columns of Table 1 carry real uncertainty even at ten seeds: the interval at 337 still includes unity. The aggregate is well separated from unity, but we make no claim about how the advantage varies with network size. The 1D control remains at three seeds, which is sufficient for a null result of its size but not for a fine reading of its columns.
- The 1D and 2D experiments differ in more than dimension: the mode count, the quadrature, the step budget and the learning rate all change between them. The steepness sweep of Section 6.3 avoids this confound entirely by varying difficulty within a single fixed setting, and it is the evidence we rely on for the mechanism; the cross-dimensional comparison should be read only as motivation.
- Goal-orientation is not uniformly beneficial: at 105 degrees of freedom and $k = 4$ it is 2.4 times worse than plain DFR, winning 2 of 10 runs. We report where it helps and where it does not, but we have no a priori test that tells a practitioner which regime a new problem is in.
- Plain DFR is not trained to convergence at the shared budget (Section 6.4), so the matched-budget figures over-credit goal-orientation. We correct for this with the matched-cost comparison of Section 6.5, giving plain DFR 2.5 times the goal-oriented budget, and rely on it: it leaves a twofold advantage in two dimensions and no reliable advantage in the steepness sweep. This matches cost, not accuracy: it still does not compare against a fully converged baseline, which these problems do not reach at practical budgets.
- Goal-oriented solutions are accurate at the quantity of interest and substantially less accurate globally, by roughly an order of magnitude in energy norm in our experiments. They should not be used where the whole field matters.
- All experiments use a single family of manufactured solutions (an arctan profile and its tensor product) with a mollified point QoI on a box. Other QoIs, other geometries and real applications remain untested.
- The goal term $|R(u_h)(z_h)|$ is non-differentiable at zero. Quasi-Newton optimizers such as L-BFGS build a Hessian approximation from successive gradients and are sensitive to this kink, so the LBFGS polish is

less effective on the goal-oriented objective than on the smooth plain-DFR loss. Two smooth surrogates with the same minimizer are natural refinements: the squared pairing $\mathcal{L}_{\text{QoI}}^{\text{sq}} = R(u_h)(z_h)^2$, and a Huber-type smoothing $\mathcal{L}_{\text{QoI}}^\varepsilon = \sqrt{R(u_h)(z_h)^2 + \varepsilon}$ that recovers $|R(u_h)(z_h)|$ as $\varepsilon \rightarrow 0$. Either would be expected to improve the quasi-Newton convergence rate; we did not require them to obtain the results above.

- The spectral evaluation of the dual norm requires a box (or box-decomposable) domain, as in the original DFR method; extending goal-orientation to re-entrant corner singularities would require pairing it with domain decomposition, as in the adaptive DFR method [17], and is left to future work.

7. Conclusion

We have presented a goal-oriented extension of the Deep Fourier Residual method. By introducing an adjoint network and minimizing a QoI-weighted dual-norm residual, the loss targets the error in a quantity of interest rather than the global energy norm, while reusing the DFR spectral machinery.

7.1. Summary

The main contributions are:

1. A goal-oriented DFR method coupling a primal and an adjoint network through a QoI-weighted dual-norm loss.
2. A goal-oriented counterpart of the DFR error-loss equivalence (Theorem 4.3): the QoI-weighted loss, plus a computable remainder, controls the error in the quantity of interest. The bound is one-sided, and it is weakest in the resolution-limited regime where goal-orientation is most useful.
3. An empirical verification of that bound, which holds on every run at a median of ten times the true error, and which shows that the QoI-weighted loss alone cannot serve as an error estimator for a network trained on it.
4. An empirical characterization of the mechanism and its size: goal-orientation reallocates accuracy from the global solution to the quantity of interest, and at equal computational cost this buys about a twofold reduction in the point-QoI error in the two-dimensional network-size setting and no reliable improvement in a sweep over problem difficulty. The theory predicts none of this.

7.2. Future work

- **A converged-baseline comparison:** the matched-cost comparison here gives plain DFR a fixed multiple of the goal-oriented budget, not a baseline trained fully to convergence, which these problems do not reach at practical budgets. Establishing the advantage against a genuinely converged baseline, or with a many-seed study that resolves the wide per-cell uncertainty, would sharpen the size estimate.
- **A bound that predicts the gain:** Theorem 4.3 is consistent with the observed advantage and does not explain it, being weakest exactly where the advantage appears. A bound that quantifies the reallocation of accuracy toward the QoI, rather than presupposing global resolution, would be the natural next result.
- **An analysis that sees the optimizer:** since a practical DFR residual reflects how far training has progressed as much as what the discretization can represent, error analysis in the best-approximation style does not capture what limits the method in practice. Bounds that account for the training dynamics are what the empirical picture calls for.
- **Nonlinear QoIs and PDEs:** extend the error representation to nonlinear functionals and operators.
- **Adaptive refinement:** combine the goal-oriented indicator with local refinement of the test space or trial networks.
- **General domains:** extend beyond box domains via domain decomposition.

7.3. Reproducibility

All code used in this paper is available in the accompanying repository, <https://github.com/caverac/hp-dfr>. The documentation site, <https://caverac.github.io/hp-dfr/>, gives instructions for reproducing the experiments and the figures.

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CRedit authorship contribution statement

Carlos Vera-Ciro: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All code, saved sweep data, and scripts to regenerate every figure are openly available in the accompanying repository, <https://github.com/caverac/hp-dfr>, with documentation at <https://caverac.github.io/hp-dfr/>.

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